

Internship Task Documentation: Titanic Dataset EDA

Exploratory Data Analysis Overview:

The goal of this analysis was to explore the Titanic dataset using visual and statistical methods to uncover patterns and insights related to passenger survival.

The analysis involved handling missing values, visualizing data distributions, examining correlations between features, and identifying factors influencing survival rates.

The steps included:

- Handling missing values:
 - Filled missing age values with the median.
 - Filled missing country values with "Unknown."
 - Dropped fare, sibsp, parch, and ticketno due to heavy missing values.
- Performing univariate and bivariate analysis:
 - Distribution of age, gender, and survival status.
 - Survival rate by gender and passenger class.
- Creating key visualizations:
 - Correlation heatmap.
 - Pairplot of important features.
 - Countplots for survival analysis.
 - Boxplot analysis based on age and survival.
- Observing relationships and trends between passenger demographics and survival.

All visualizations were built using a consistent color theme for better readability and understanding.

Key Findings:

Gender Distribution and Survival

- A significantly higher proportion of females survived compared to males.
- Gender played a major role in survival chances.

Survival by Passenger Class

- Passengers in 1st Class had the highest survival rate.
- 3rd Class passengers had the lowest survival rates.
- Socioeconomic status was a major factor influencing survival.

Age and Survival

- Younger passengers, especially children, had better survival chances.
- Older passengers had a lower probability of survival.
- Boxplot analysis showed a clear age pattern between survivors and non-survivors.

Correlation Analysis

- Moderate correlation observed between sibsp and parch (family travel groups).
- Age had slight variations with other features but no very strong correlation.

Conclusion & Recommendations:

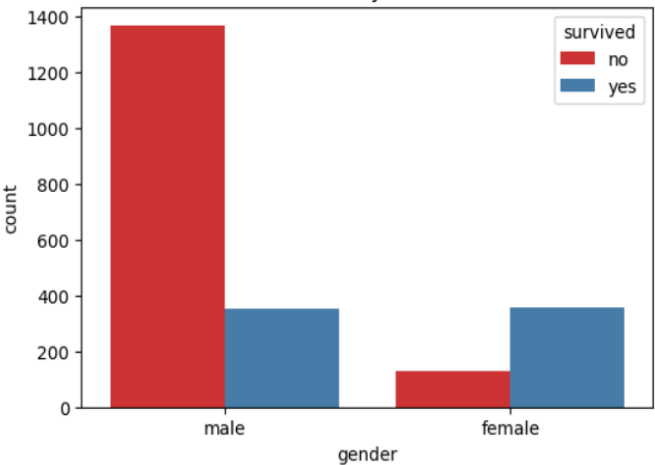
The exploratory analysis successfully identified key factors affecting survival on the Titanic:

- Gender and class were primary influences on survival chances.
- Younger passengers had a higher likelihood of survival.
- Family connections (siblings, parents) might have influenced survival behaviour, but further detailed analysis would be needed.

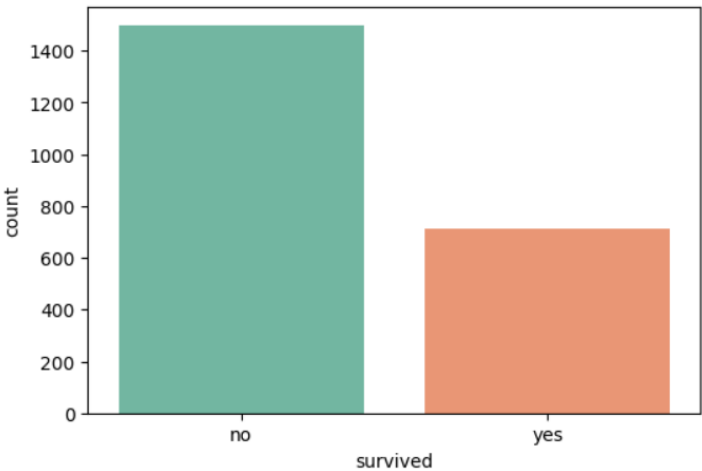
It is recommended that in further modeling stages (like predictive modeling), variables such as gender, class, and age should be given higher importance

Reports:

Survival by Gender



Survival Distribution



Survival by Class

